Agisoft PhotoScan

Processing Report

24 September 2015

Survey Data

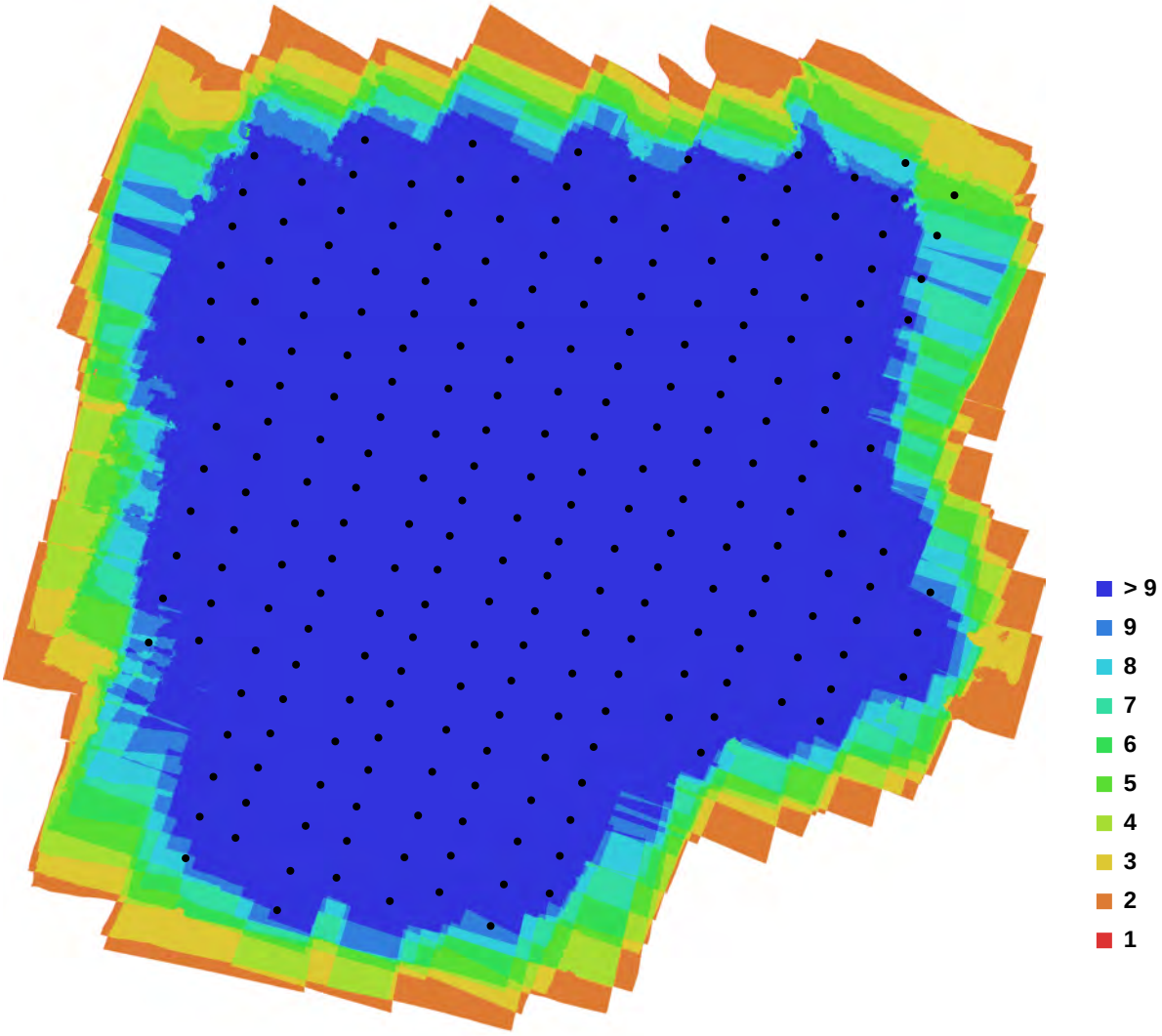

Fig. 1. Camera locations and image overlap.

Number of images:	360	Camera stations:	244
Flying altitude:	116.645 m	Tie-points:	23920
Ground resolution:	0.0316475 m/pix	Projections:	243822
Coverage area:	0.306662 sq km	Error:	1.61011 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
NEX-5T	4912 x 3264	unknown	unknown	No

Table. 1. Cameras.

Camera Calibration

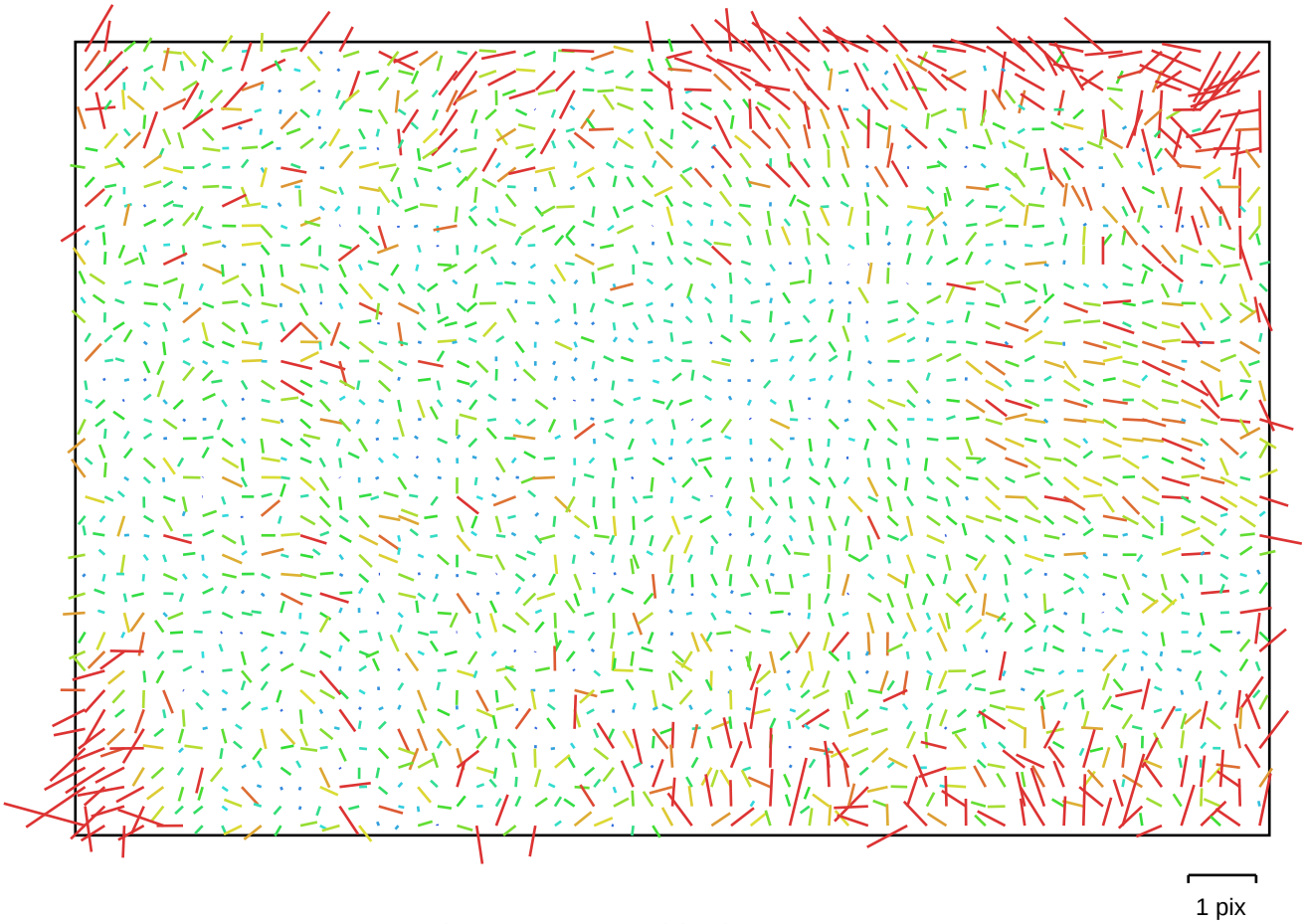

Fig. 2. Image residuals for NEX-5T.

NEX-5T

Type:	Frame	K1:	-0.0521908
Fx:	3325.17	K2:	0.0451292
Fy:	3322.44	K3:	-0.0149568
Cx:	2491.55	K4:	0
Cy:	1679.45	P1:	-0.00114318
Skew:	-35.7725	P2:	0.000584998

Camera Locations

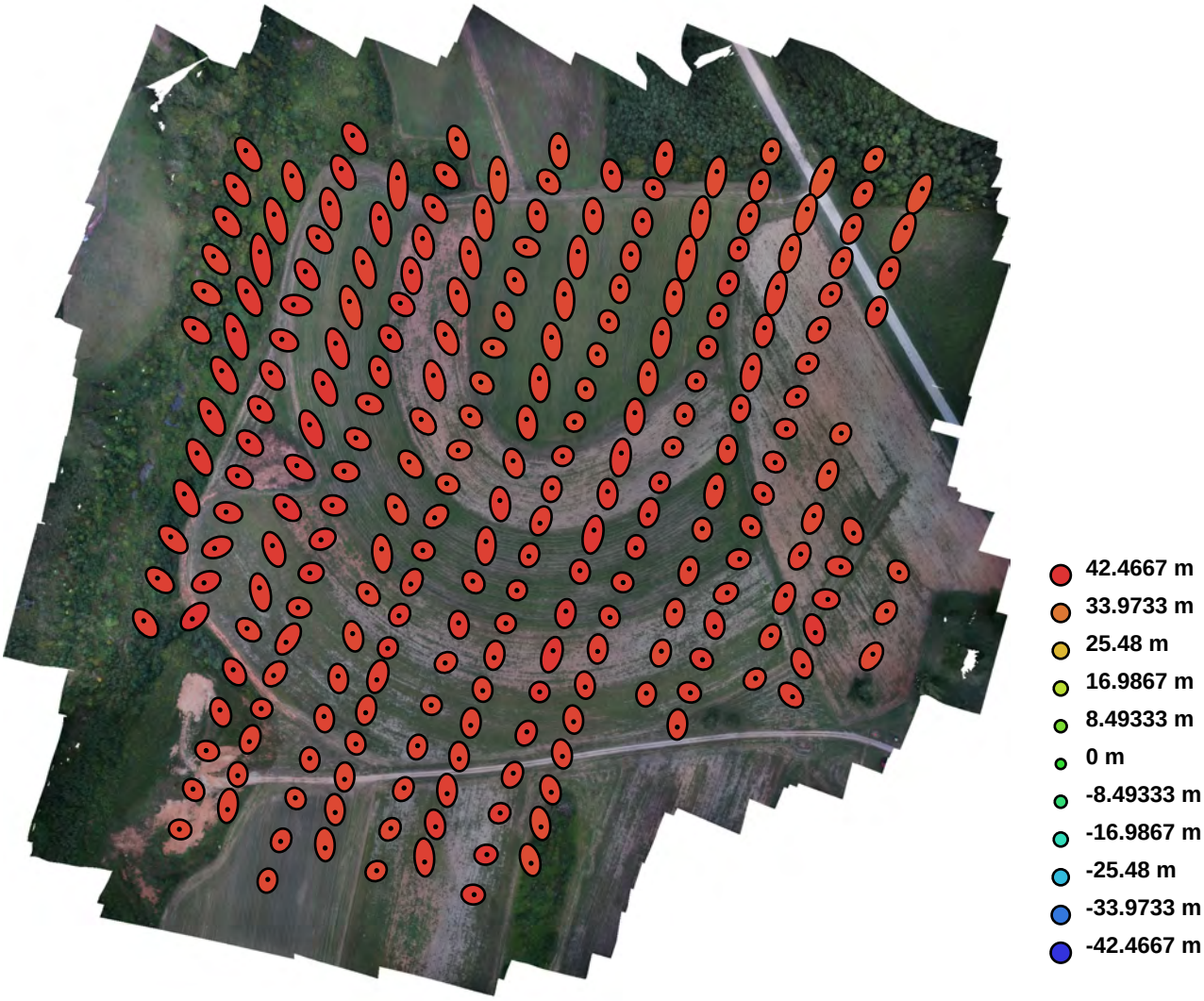

Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	Total error (m)
1.594437	3.794176	40.340008	40.549405

Table. 2. Average camera location error.

Ground Control Points

Fig. 4. GCP locations.

Label	X error (m)	Y error (m)	Z error (m)	Error (m)	Projections	Error (pix)
UAV 1	-0.000000	0.000000	-0.000000	0.000000		
UAV 10	-0.035362	-0.114797	-0.114315	0.165821	3	0.303362
UAV 11	-0.000000	-0.000000	0.000000	0.000000		
UAV 12	0.000000	0.000000	0.000000	0.000000		
UAV 2	-0.000000	0.000000	-0.000000	0.000000		
UAV 3	0.000000	0.000000	-0.000000	0.000000		
UAV 4	-0.000000	0.000000	0.000000	0.000000		
UAV 5	0.134997	-0.167779	-0.000823	0.215348	3	0.090671
UAV 6	-0.000000	0.000000	-0.000000	0.000000		
UAV 7	-0.000000	0.000000	0.000000	0.000000		

Label	X error (m)	Y error (m)	Z error (m)	Error (m)	Projections	Error (pix)
UAV 8	0.115056	0.069636	-0.015976	0.135434	3	0.299831
UAV 9	-0.246624	0.174716	-0.000581	0.302240	5	0.251384
Total	**0.088287**	**0.079949**	**0.033322**	**0.123681**	**14**	**0.251625**

Table. 3. Control points.

Digital Elevation Model

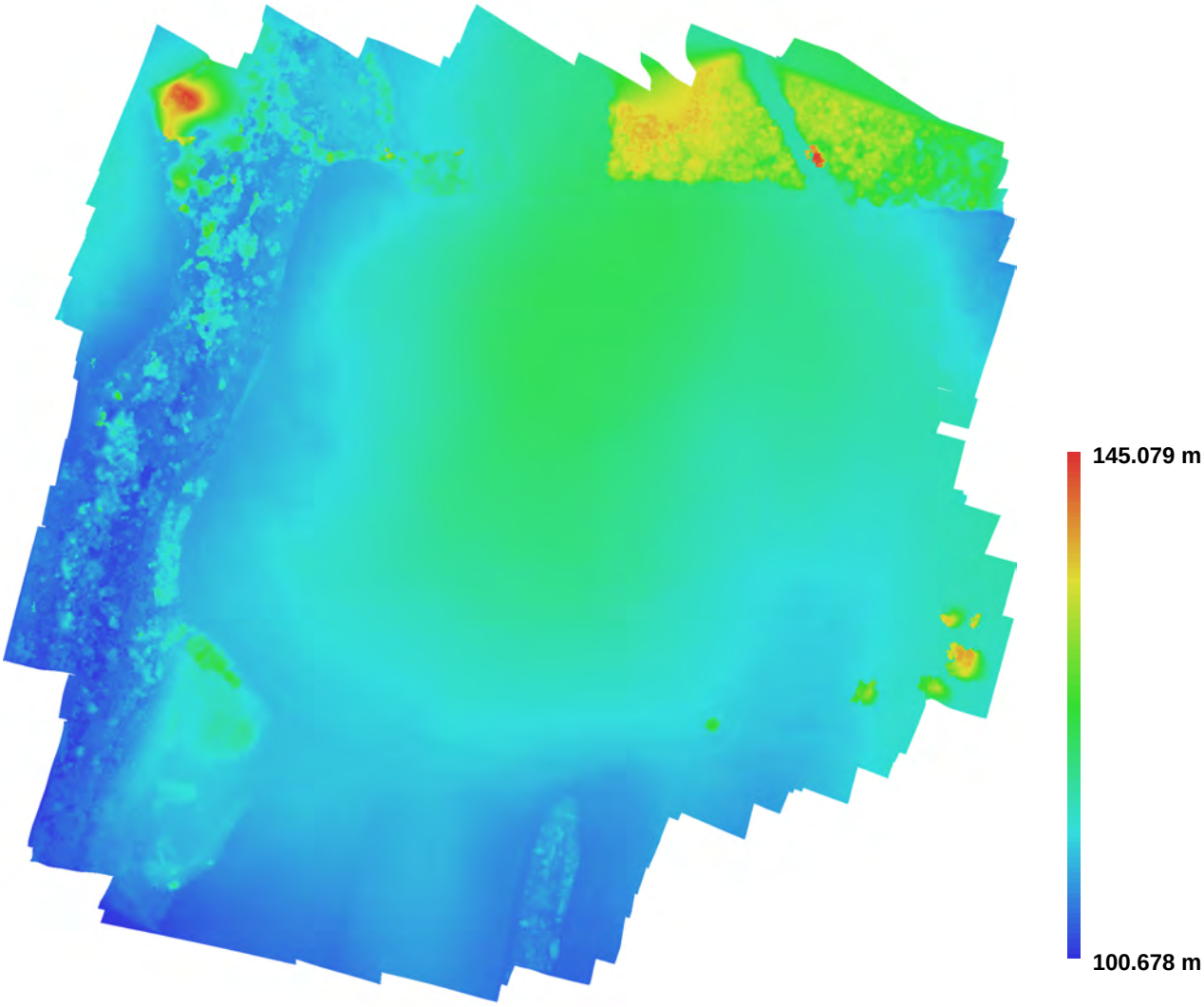

Fig. 5. Reconstructed digital elevation model.

Resolution:	0.0632949 m/pix
Point density:	249.61 points per sq m